

Multilingual Government Scheme Information Retrieval System

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Abstract—It is a significant challenge for Indian citizens who are not proficient in English to access information about government schemes because India is a linguistically diverse region, and some Indian citizens might be well-versed in Indian local languages rather than English. This project is the design and development of a multilingual and multimodal government scheme information retrieval system that supports English, Hindi, and Kannada. This system makes it accessible to non-literate users to interact using both speech and text inputs. A multilingual embedding model is used to generate semantic representations of user queries and schemes, allowing effective retrieval through similarity search over a vector database. Unlike keyword-based methods, this system uses semantic search to be more accurate and provide context-aware results. It also supports cross-lingual retrieval, where queries in one language will fetch relevant documents in another. The information that was retrieved is translated into the user's preferred language, and it is displayed in text format. This ensures ease of understanding and usability. By integrating Natural Language Processing, Speech Processing, and Cross - Lingual capabilities, we aim to reduce information division caused by linguistic diversity. In a nutshell, this system ensures inclusive digital access and provides more effective communication to the public.

I. INTRODUCTION

Accessing welfare schemes from the government is quite challenging due to the lack of English proficiency and familiarity with search mechanisms among several individuals. Existing portals are mainly text-based and require users to use keywords when searching. In addition, the portal is based on digital information technology, which may not be suitable for everyone, as most applications are mainly designed for English, whereas many people in India are more familiar with regional languages. Hence, the

majority of citizens fail to learn about government welfare schemes.

Recently, advancements in natural language processing, speech processing, and semantic information retrieval techniques have provided the basis for the implementation of intelligent multilingual information search systems. A standard keyword search engine responds to a query on the presence of keywords in a query, whereas a semantic search engine responds to the actual meaning of queries. Semantic search engines can thus provide good results regarding the government schemes since queries are usually made using informal language. Moreover, cross-lingual information retrieval allows a query made in one language to retrieve data in another language, such as from Kannada to English.

Other than retrieval, in designing a citizen-centric system, the results must be obtained in the same language as the query. It is essential to develop a system that would allow the results to be returned in the form of text.

The proposed study will focus on the development of a system for accessing information on government schemes in three languages: Kannada, Hindi, and English. The system accepts queries from users in text as well as speech format, and a multilingual embeddings model is used to embed queries and schemes into a common semantic space. Information retrieval takes place based on similarity searches in vector databases.

The main objective of the study is to improve accessibility and availability of information to regional language speakers and low literacy users by implementing cross-lingual information retrieval in multiple Indian languages.

II. RELATED WORK

A. Retrieval-Based Multilingual Systems

Recent research in information retrieval and cross-lingual document retrieval focused on improving retrieval across multilingual and low-resource settings. Frameworks such as PRALEKHA[1] address cross-lingual document alignment by building large-scale multilingual datasets and initiating embedding-based similarity metrics that add value to models like LaBSE and SONAR for aligning small parts of documents. This demonstrates improved performance when compared to traditional methods. Likewise, systems like IndicIRSuite[2] introduce large-scale multilingual retrieval benchmarks and neural retrieval models, such as Indic-ColBERT, both showcasing the advantages of dense semantic retrieval over traditional approaches like BM25, especially in Indian languages. In contrast, approaches that are more classical, like Latent Semantic Analysis (LSA)[3], explore multilingual retrieval through dimensionality reduction techniques applied to TF-IDF representations. This captures latent semantic relationships across languages. While the neural and embedding-based methods significantly outperform traditional techniques regarding semantic understanding and retrieval accuracy, challenges remain in handling low-resource languages, domain-specific datasets, and highly paraphrased or noisy text, which affects retrieval performance and generalization. These challenges need to be resolved for our project context, as we are including lingual diversity, and we are also using a domain-specific dataset.

B. Retrieval-Augmented Generation (RAG)-Based Systems

The formation of Retrieval-Augmented Generation (RAG) based applications has improved multilingual understanding and domain-specific information retrieval. Approaches such as mRAKL[4] have redefined knowledge graph completion as a system that is used for question-answering, which provides cross-lingual knowledge inference through multilingual retrieval and generation. Multilingual RAG[5] integrates dense-retrieval with sequence-to-sequence models by embedding queries and documents into a shared vector space. This helps in the collection of relevant context for response generation. Frameworks like XRAG[6] and IndicRAGSuite[7] highlight multilingual retrieval challenges, like maintaining response language consistency and low-resource languages. In parallel, domain-specific systems such as LegalRAG[8] show the advantage of hybrid retrieval methods in legal document analysis, but they are limited due to limited datasets and insufficient domain adoption. While RAG-based systems show notable progress in improving retrieval accuracy and generating responses across diverse languages and fields, their overall impact is dependent on retrieval quality, multilingual resources, and robustness of linguistic variations.

C. Retrieval-Augmented Generation (RAG)-Based Systems

The current state of research in multimodal and application-based NLP systems focuses primarily on enhancing access to information and cross-modal processing capabilities. The work on LocatingGPT[9] involves

extending the RAG architecture to accommodate multimodal inputs through the use of Donut for document images and Whisper for transcribing audio inputs, alongside deep semantic parsing to increase retrieval effectiveness. Moreover, dual-encoder-based solutions, like PaLM2DE[10], allow transforming language models to cross-modal solutions by mapping representations of speech and text in a unified embedding space and facilitating efficient cross-lingual and cross-modal retrieval of knowledge. Concurrently, there are applications of NLP-based pipelines[11][12] designed to increase accessibility of information in particular applications, such as governance processes. This refers to the development of multilingual pipelines that allow simplifying, translating, and providing semantic enrichment of governmental documents and other documents of a complex nature through the conversion of policies into audio-visual documents via text simplification, sentiment analysis, and generation. Nevertheless, while these systems significantly improve the readability and accessibility of information as well as the retrieval process across different languages and modalities, they still require improvements in terms of efficiency and computational complexity and rely on high-quality pretrained models.

III. METHODOLOGY

A. System Overview

The proposed system is a multilingual retrieval system that aims to make access easier to welfare schemes. It takes queries in textual or spoken formats in English, Kannada, and Hindi. Language detection and query processing were done before retrieving relevant schemes. A hybrid retrieval system was proposed that uses semantic search, BM25 keyword match, and fuzzy match techniques. The top-k schemes are then passed into Retrieval-Augmented Generation for response generation

B. Data Collection and Preprocessing

The database that we used in this project is “Indian Government Schemes” from Kaggle. This database is compiled through the process of web scraping all central and state schemes in India available on the official website of the government, “MyScheme”. We developed a multilingual database by transliterating the names of the schemes into Kannada and Hindi languages using Google's Transliteration. Tags associated with the schemes were translated into Kannada language (Google Trans) and Hindi language (Marinatransformer). This preserves semantic identity and ensures consistent recognition across languages. In contrast, tags represent categories and are translated to improve semantic understanding. Scheme names and tags in all three languages, along with other details, are combined into a

single string. This multilingual string is stored as a separate column and is converted to dense numerical vectors/embeddings.

C. Multilingual Query Processing

Users can either give their query in text or speech form in English, Hindi, or Kannada.

a. Language Identification

Each language has its own Unicode range, and this Unicode range comparison is used to determine the language of the query. It helps in handling multilingual queries without depending on any external API. The language is identified as follows:

Lang(q) = { Kannada, if q ∈ [U+0C80 to U+0CFF]
Hindi, if q ∈ [U+0900 to U+097F]
English, otherwise }

b. Query Cleaning and Tokenization

The user query is then normalized by removing whitespaces, handling punctuation, and lowercasing for English text. Next, tokens are extracted for keyword-based retrieval. This query is presented as :

$$q = \{w_1, w_2, w_3, \dots, w_n\} \quad (1)$$

where w_i represents individual tokens

D. Document Retrieval Module

A hybrid retrieval strategy is used, which combines semantic vector retrieval, keyword retrieval, and fuzzy string match. This helps in handling spelling variations and retrieving exact scheme names.

a. Fuzzy Name Matching

The input query might contain spelling errors or partial words. This is handled by performing fuzzy matching using RapidFuzz on the scheme name column and retaining only the matches above a minimum threshold. The fuzzy similarity between the query q and scheme name d_i is computed as:

$$F(q, d_i) = \text{TokenSortRatio}(q, d_i) \quad (2)$$

b. Semantic Retrieval using Sentence Embeddings and FAISS

Each multilingual string S_i is converted into a dense vector

representation using a SentenceTransformer model, paraphrase-multilingual-MiniLM-L12-v2. This model supports cross-lingual semantic understanding, which means that a query in one language can retrieve documents in another language.

$$e_i = f_{\text{embed}}(S_i) \quad (3)$$

where $e_i \in \mathbf{R}^d$ is the embedding of document i

And f_{embed} is a multilingual embedding model

Next, these embeddings are normalized using L2 normalization, which improves cosine similarity search efficiency:

$$\hat{e}_i = \frac{e_i}{\|e_i\|} \quad (4)$$

Similarly, query embedding is computed and normalized.

$$\hat{q} = \frac{f_{\text{embed}}(q)}{\|f_{\text{embed}}(q)\|} \quad (5)$$

After normalization, the similarity between a query and a document is calculated using the inner product:

$$\text{Sim}_{\text{semantic}}(q, d_i) = \hat{q} \cdot \hat{e}_i \quad (6)$$

To handle large-scale document collections, all embeddings are indexed using Facebook AI Similarity Search (FAISS). This enables nearest-neighbor search, allowing one to find the most relevant document for a given query quickly.

c. Keyword Retrieval using BM25

BM25, a probabilistic ranking algorithm designed for keyword - based document retrieval, is also added, which is efficient when the user query contains exact domain-specific words like farmer, scholarship, etc.

For a query q and document d , BM25 is computed as

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d) \cdot (k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \quad (7)$$

where $f(t, d)$ is the frequency of term t in document d

$|d|$ is document length

Avgdl is the average document length

K_i and b are tuning constants

d. Hybrid Ranking

The final retrieval score is calculated by combining semantic similarity and BM25 relevance. The hybrid score for a document d_i is computed as:

$$Score(q, d_i) = \alpha \cdot Sim_{semantic}(q, d_i) + \beta \cdot BM25(q, d_i) \quad (8)$$

where α (semantic weight) is 0.7, β (BM25 weight) is 0.3, and $\alpha + \beta = 1$

This weighted combination improves retrieval quality by balancing semantic understanding and keyword precision. Next, the top-k most relevant documents are returned based on the combined score:

$$R_k = TopK(Score(q, d_i)) \quad (9)$$

where q is the query

d_i represents each document

$Score(q, d_i)$ is the combined (semantic + BM25) score

$TopK$ selects the top k (5) highest-scoring documents

E. Retrieval-Augmented Generation

The system fetches the top 5 schemes from its search using RAG (Retrieval-Augmented Generation) to generate more accurate and grounded responses. Let the retrieved document set be

$$R_k = \{d_1, d_2, \dots, d_k\} \quad (10)$$

The model generates answers based on user queries q and the context R_k :

$$A = G(q, R_k) \quad (11)$$

where A represents the generated answer, and G denotes the response generation model.

This ensures that the generated response is grounded and retrieved from actual data instead of relying solely on model memory. The document provides information like scheme title, eligibility criteria, benefits, category, official portal link, and implementation level.

F. Response Generation and Post-Processing

After extracting the schemes, the output is generated in the form of an organised response. Each response contains

scheme names in English, Hindi, and Kannada, scheme category, level, related tags, and a unique link for the official scheme page via slug-based URLs. The response is then post-processed to ensure multilingual readability, result ranking, and formatting for readability.

G. Speech Integration

In order to make the system more accessible for users who might not have typing capabilities or reading skills, speech queries are enabled using a speech-to-text converter.

a. Speech Recording and Pre-Processing

Voice queries are captured via the browser microphone and stored as audio files. To facilitate accurate speech detection, the audio is preprocessed as a monaural WAV file at 16 kHz. This conversion is necessary to ensure compatibility and recognition quality.

Let the recorded audio signal be $x(t)$.

The preprocessing stage converts it into a normalized waveform:

$$x'(t) = Preprocess(x(t)) \quad (12)$$

where preprocessing includes resampling, channel reduction, and noise handling.

b. Speech-to-text using whisper

The processed audio is transcribed by OpenAI Whisper, taking into consideration the medium model, which supports Indian languages like Hindi and Kannada well.

The transcription is given by: $\hat{x}$

$$q = STT(x'(t)) \quad (13)$$

Where $x'(t)$ represents processed audio input, q is the transcribed text query, and STT denotes the speech-to-text model.

We also allow users to optionally pass a language hint like Hindi, Kannada, or English, which can aid in providing better results, especially for multilingual and code-mixed speech. The obtained results are then passed into a hybrid retrieval pipeline, thereby enabling a unified architecture for text-based and voice-based access.

IV. RESULTS AND DISCUSSION

This project was evaluated using a set of 19 queries across three different languages: English, Hindi, and Kannada. The evaluation focuses on measuring the effectiveness of the

retrieval system using Hit Rate@K, Mean Precision@K, and Mean Reciprocal Rank (MRR).

The system was able to achieve Hit Rate@5 of 94.7%, indicating that for most of the queries, most likely at least one relevant scheme will be successfully retrieved within the top 5 results. The Mean Precision@5 was 86.3%, demonstrating good relevance for a high proportion of retrieved results. The MRR of 0.9474 shows that relevant scheme results were typically ranked very high (often at 1st place).

Language-wise performance

The system was evaluated for English, Hindi, and Kannada queries to assess multilingual effectiveness.

English queries achieved a Hit Rate@5 of 100% and Mean Precision of 94.3%, indicating very good performance as it aligns almost perfectly with the dataset.

Hindi queries also achieved a Hit Rate@5 of 100% and Mean Precision of 96.7%, which shows that cross-lingual retrieval capabilities are very effective.

Kannada queries showed slightly lower performance when compared to both English and Hindi, with a Hit Rate@5 of 83.3% and Mean Precision of 66.7%, and this is mostly due to morphological complexity and models having less proficiency in Kannada.

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REFERENCES

- [1] S. Suryanarayanan, H. Song, M. S. U. R. Khan, A. Kunchukuttan, and R. Dabre, "PRALEKHA: Cross-lingual document alignment for Indic languages," 2025.
- [2] S. Haq, A. Sharma, O. Khattab, N. Chhaya, and P. Bhattacharyya, "IndicIRSuite: Multilingual dataset and neural information models for Indian languages," 2024.
- [3] P. Sethuraman, M. M. Hassan, R. P. Veetil, A. Jayanthi, N. Kalyana Sundaram, and D. Patnaik, "Evaluation of latent semantic analysis in multilingual information retrieval," 2025.
- [4] H. H. Nigatu, M. Li, M. ter Hoeve, S. Potdar, and S. E. Chasins, "mRAKL: Multilingual retrieval-augmented knowledge graph construction for low-resourced languages," 2025.
- [5] L. Ranaldi, B. Haddow, and A. Birch, "Multilingual retrieval-augmented generation for knowledge-intensive tasks," 2025.
- [6] W. Liu, S. Trenous, L. F. R. Ribeiro, B. Byrne, and F. Hieber, "XRAG: Cross-lingual retrieval-augmented generation," 2025.
- [7] P. Prasanjith, P. B. More, A. Kunchukuttan, and R. Dabre, "IndicRAGSuite: Large scale datasets and a benchmark for Indian language RAG systems," 2025.
- [8] M. R. Kabir, R. M. Sultan, F. Rahman, M. R. Amin, S. Momen, N. Mohammed, and S. Rahman, "LegalRAG: A hybrid RAG system for multilingual legal information retrieval," 2025.
- [9] Z. Chen, P. Zhang, M. Xu, X. Gao, D. Liu, X. Luo, R. Huang, and T. Zhang, "LocatingGPT: A multi-modal document retrieval method based on retrieval-augmented generation," 2024.
- [10] F. Palma Gomez, R. Sanabria, Y.-H. Sung, D. Cer, S. Dalmia, and G. Hernandez Abrego, "Transforming LLMs into cross-modal and cross-lingual retrieval systems," 2024.
- [11] M. Maulana and J. Estrada, "A multilingual NLP pipeline for semantic enrichment and search of local government ordinances: A case study from Davao Oriental," 2025.
- [12] K. T., P. Ramadoss, M. M., and M. S. S., "A multilingual framework based on NLP for disseminating government policies via audio-visual content," 2025.